Lab 5 — Coverage

FIFO

Task 1

Question 1.1: Covergroup Bins

The testbench defines one `covergroup` (`fifo_cov`) and one `cover property` directive. In total there are **13 covergroup bins + 1 cover directive**.

Coverpoint `full` (1 bin)

1. `went_full` — bin for `full == 1`

Coverpoint `empty` (1 bin)

2. `went_empty` — bin for `empty == 1`

Coverpoint `dut.count` (3 bins)

3. `empty_bin` — occupancy equals 0
4. `mid_bins` — occupancy in range `[1 : DEPTH-1]`
5. `full_bin` — occupancy equals `DEPTH`

Coverpoint `wr_en` (2 auto-bins)

6. `auto[0]` — `wr_en == 0`
7. `auto[1]` — `wr_en == 1`

Coverpoint `rd_en` (2 auto-bins)

8. `auto[0]` — `rd_en == 0`
9. `auto[1]` — `rd_en == 1`

Cross `wr_en` × `rd_en` (4 bins)

10. `write_only` — `wr_en=1, rd_en=0`
11. `read_only` — `wr_en=0, rd_en=1`
12. `both_off` — `wr_en=0, rd_en=0`
13. `illegal` — `wr_en=1, rd_en=1`

Cover Directive (+1)

D1. `cover property (becomes_empty_after_read)` — covers the sequence where the FIFO transitions to empty after a read when `dut.count > 0`.

Question 1.2: Bins Covered After Reset Only

The following **5 bins** are covered just from reset:

1. `went_empty` — after reset the FIFO is empty, so `empty == 1`
2. `empty_bin` — `dut.count == 0` after reset
3. `wr_en auto[0]` — `wr_en` stays 0
4. `rd_en auto[0]` — `rd_en` stays 0
5. `both_off` — cross bin where `wr_en=0, rd_en=0`

Task 2

Question 2.1: Additionally Covered Bins

After enabling the provided stimulus, the following **5 additional bins** are now covered:

1. `write_only` — cross bin where `wr_en=1, rd_en=0`
2. `read_only` — cross bin where `wr_en=0, rd_en=1`
3. `wr_en auto[1]` — `wr_en == 1`
4. `rd_en auto[1]` — `rd_en == 1`
5. `mid_bins` — occupancy in range `[1 : DEPTH-1]`

Task 3

Missing Bins After Task 2

The following **3 bins + 1 cover directive** remained uncovered:

1. `went_full` — `full == 1`
2. `full_bin` — `dut.count == DEPTH`
3. `illegal` — cross bin where `wr_en=1, rd_en=1`

D1. `becomes_empty_after_read` — FIFO transitions to empty after a read when `dut.count > 0`

Additional Stimulus

Fully fill and fully empty the FIFO. Increased the number of consecutive writes to exceed `DEPTH` (16), driving the FIFO to full occupancy, then increased the number of consecutive reads to drain it completely back to empty.

Hits bins:

- `went_full` — FIFO reaches full capacity
- `full_bin` — `dut.count == DEPTH`
- `becomes_empty_after_read` (D1) — FIFO drains to empty after a read while `dut.count > 0`

Drive both `wr_en` and `rd_en` on the same cycle. Added stimulus that asserts `wr_en=1` and `rd_en=1` simultaneously.

Hits bin:

- `illegal` — cross bin where `wr_en=1, rd_en=1`

Cache

Task 4

Covergroup Bins

The testbench defines one `covergroup` (`cache_cov`) with **9 coverpoints/crosses** totalling **27 bins**, plus **7 cover property** directives.

Coverpoint `write` (2 auto-bins)

1. `auto[0]` — `write == 0`
2. `auto[1]` — `write == 1`

Coverpoint `read` (2 auto-bins)

3. `auto[0]` — `read == 0`
4. `auto[1]` — `read == 1`

Coverpoint `hit` (2 auto-bins)

5. `auto[0]` — `hit == 0`
6. `auto[1]` — `hit == 1`

Cross `write` × `read` (4 bins)

7. `write_only` — `write=1, read=0`
8. `read_only` — `write=0, read=1`
9. `both_off` — `write=0, read=0`
10. `read_over_write` — `write=1, read=1`

Cross `write` × `hit` (4 bins)

11. `write_hit` — `write=1, hit=1`
12. `write_miss` — `write=1, hit=0`
13. `auto[write=0][hit=1]` — not writing, hit asserted (carried from prior op)
14. `auto[write=0][hit=0]` — not writing, no hit

Cross `read` × `hit` (4 bins)

15. `read_hit` — `read=1, hit=1`
16. `read_miss` — `read=1, hit=0`
17. `auto[read=0][hit=1]` — not reading, hit asserted (carried from prior op)
18. `auto[read=0][hit=0]` — not reading, no hit

Coverpoint `dut.tag` (3 bins)

19. `lower_bins` — tag in range `[0 : TAG_LO]`
20. `mid_bins` — tag in range `[TAG_LO+1 : TAG_HI-1]`
21. `high_bins` — tag in range `[TAG_HI : TAG_MAX]`

Coverpoint `dut.index` (3 bins)

22. `lower_bins` — index in range `[0 : INDEX_LO]`
23. `mid_bins` — index in range `[INDEX_LO+1 : INDEX_HI-1]`
24. `high_bins` — index in range `[INDEX_HI : INDEX_MAX]`

Coverpoint `dut.offset` (3 bins)

25. `lower_bins` — offset in range `[0 : OFFSET_LO]`
26. `mid_bins` — offset in range `[OFFSET_LO+1 : OFFSET_HI-1]`
27. `high_bins` — offset in range `[OFFSET_HI : OFFSET_MAX]`

Cover Directives (7)

D1. `p_read_replace` — read to a valid line with tag mismatch (conflict miss / replacement)

D2. `p_write_replace` — write to a valid line with tag mismatch (conflict miss / replacement)

D3. `p_read_cold_miss` — read to an invalid line (compulsory miss)

D4. `p_write_cold_miss` — write to an invalid line (compulsory miss)

D5. `p_write_then_read_hit` — write followed by read hit to the same address within 5 cycles (data round-trip)

D6. `p_read_miss_then_hit` — read miss followed by read hit to the same address within 5 cycles (cache fill path)

D7. `p_hit_deasserts` — hit transitions $1 \rightarrow 0$ (verifies hit is not stuck high)

Bins Covered by Initial Random Test

26 of 27 covergroup bins and 5 of 7 cover directives were hit. The following 3 were not:

- `read_over_write` (cross `write` × `read`)
- `p_write_then_read_hit`
- `p_read_miss_then_hit`

Task 5

Stimulus changes made to achieve 100

1. Weighted constrained-random operation class. Replaced the 50/50 `$urandom_range` with a SystemVerilog transaction class using a `dist` constraint, enabling all four operation combinations to be generated:

Operation	Weight	Signals
Read only	45%	`read=1, write=0`
Write only	45%	`read=0, write=1`
Simultaneous	5%	`read=1, write=1`
Idle	5%	`read=0, write=0`

Covers: `read_over_write` — the 5% simultaneous weight guarantees `read=1, write=1` is generated within 200 cycles.

2. Directed sequences for same-address access. Two directed loops were added after the random phase:

- *Write then read-hit:* writes to a random address, then reads the same address on the next cycle. Repeated 20 times. Covers `p_write_then_read_hit`.
- *Read-miss then read-hit:* reads a random address (miss), then reads the same address on the next cycle (guaranteed hit). Repeated 20 times. Covers `p_read_miss_then_hit`.